Part A. Presentation and Justification of the Problem

Problem

The task of this project is to design, build, and test a water filter that can provide clean drinking water to people that do not have access to it. A lot of people take having clean drinking water for granted since they can just turn on their sink and clean water comes out. In other places however, they do not have a sink or anything they can turn on and clean water comes out. So they are left with a lot of dirty water that is unsafe and might have to travel far to get water to drink. Designing a water filter for them to use to filter the dirty water and make it safe to drink would make life a lot easier for them to have access to clean drinking water. They also have to have a way to purify the water as well after filtering it. This can be done by boiling the water or using something to remove the bacteria and other contaminants.

Justification

Around two billion people around the world do not have access to clean drinking water. 3.6 billion people which is 46% of the world's population lack aquatic sanitation services so they don't have access to water that is safe to drink. Their water has parasites, bacteria, and other diseases that can cause them to get sick and not have good health. It is important to have clean drinking water since it is something that we need to survive.